



磷酸铁锂工作汇报



报告人: xxx 时 间: 2025年11月xx日

01 | 文献自动化获取

02 | 结构化信息提取

03 | 多智能体问答系统的搭建

04 | 问答系统流程

05 | 后续计划

01 | 文献自动化获取



01 文献自动化获取

——获取LFP相关文献



中国知网



爬虫脚本

Python + Playwright + Headless浏览器控制

自动化登录校园图书馆系统
跨浏览器支持
检索词、下载数量精确控制
防反爬机制

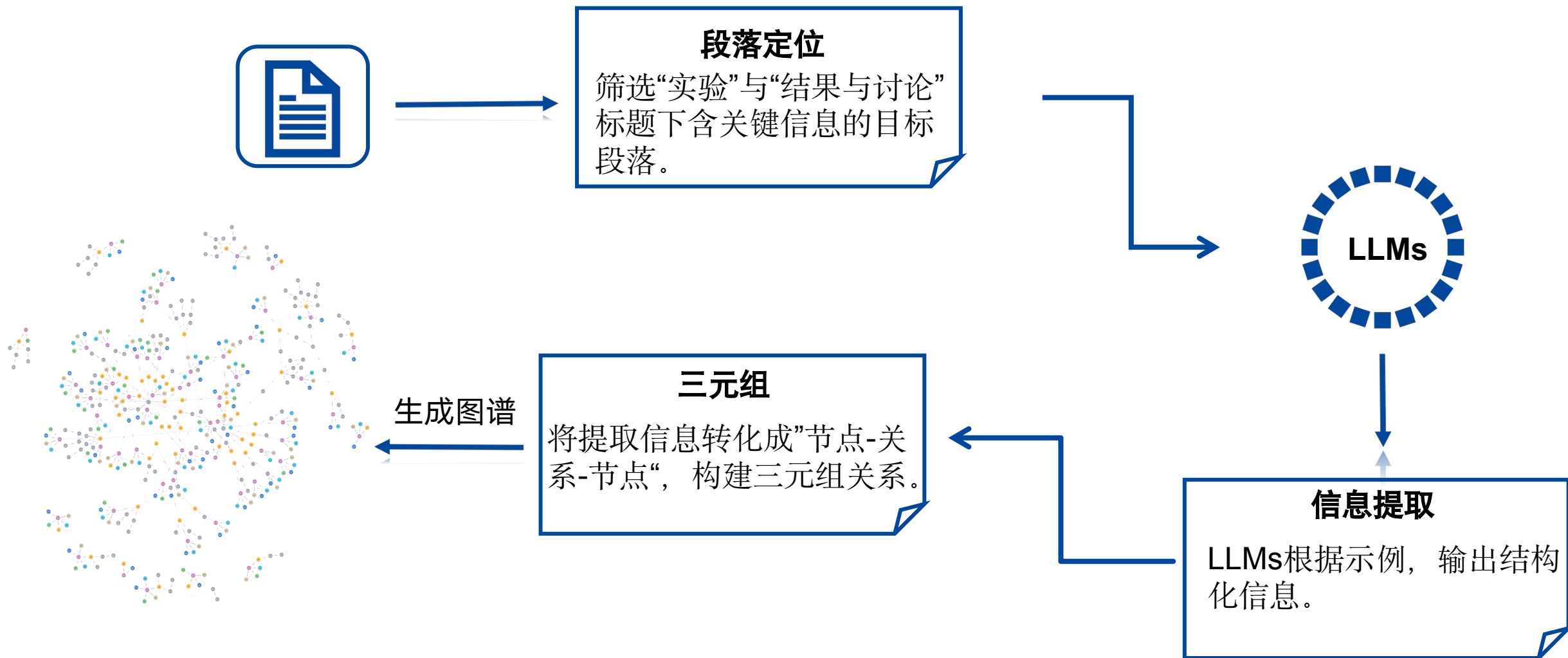
下载英文文献15820篇,
中文文献1914篇。

02| 结构化信息提取



02 结构化信息提取

——信息提取任务



——文献信息提取

一级	二级
原料	名称、生产厂家、纯度
配方	铁磷比、锂铁比、元素掺杂、碳源、添加剂
工艺	砂磨喷雾粉碎参数、烧结制度、工艺流程、工艺路线
性能	压实密度、形貌（圆润度）、碳含量、粉末电阻率、粒度、容量、能效
设备	名称、型号、构造、参数、产能
测试	测试方案、仪器名称

按照上述表格要求，设置规则与prompt对文献进行批量信息提取。以右侧获取的配方数据为例，不同文献对于比例的描述不一致，但LLMs能够根据不同的描述输出准确的内容。

2.2. Synthesis of nanocomposite

A set of $\text{Li}_{1-x}\text{Fe}_x\text{PO}_4$ were synthesized by microwave-hydrothermal method by different the ferric and lithium ratio as ($x = 0.2, 0.4, 0.6$ and 0.8). In a procedure, (0.2 M, 20 ml) of each ferric and lithium chloride were prepared and mixed together until got yellow clear solution. Then, (0.1 M, 10 ml) Na_2HPO_4 was added dropwise to mixture under strong stirring. After that, the mixture was transferred to autoclave and subjected it under $100\text{ }^\circ\text{C}$ for 100 min at rate of heating $10\text{ }^\circ\text{C}$ per min. The light brown colloidal solution was centrifuged and washed three times by distilled water before dried it at $80\text{ }^\circ\text{C}$ for 3 h.

10.1016/j.ijoes.2025.101075

LLMs

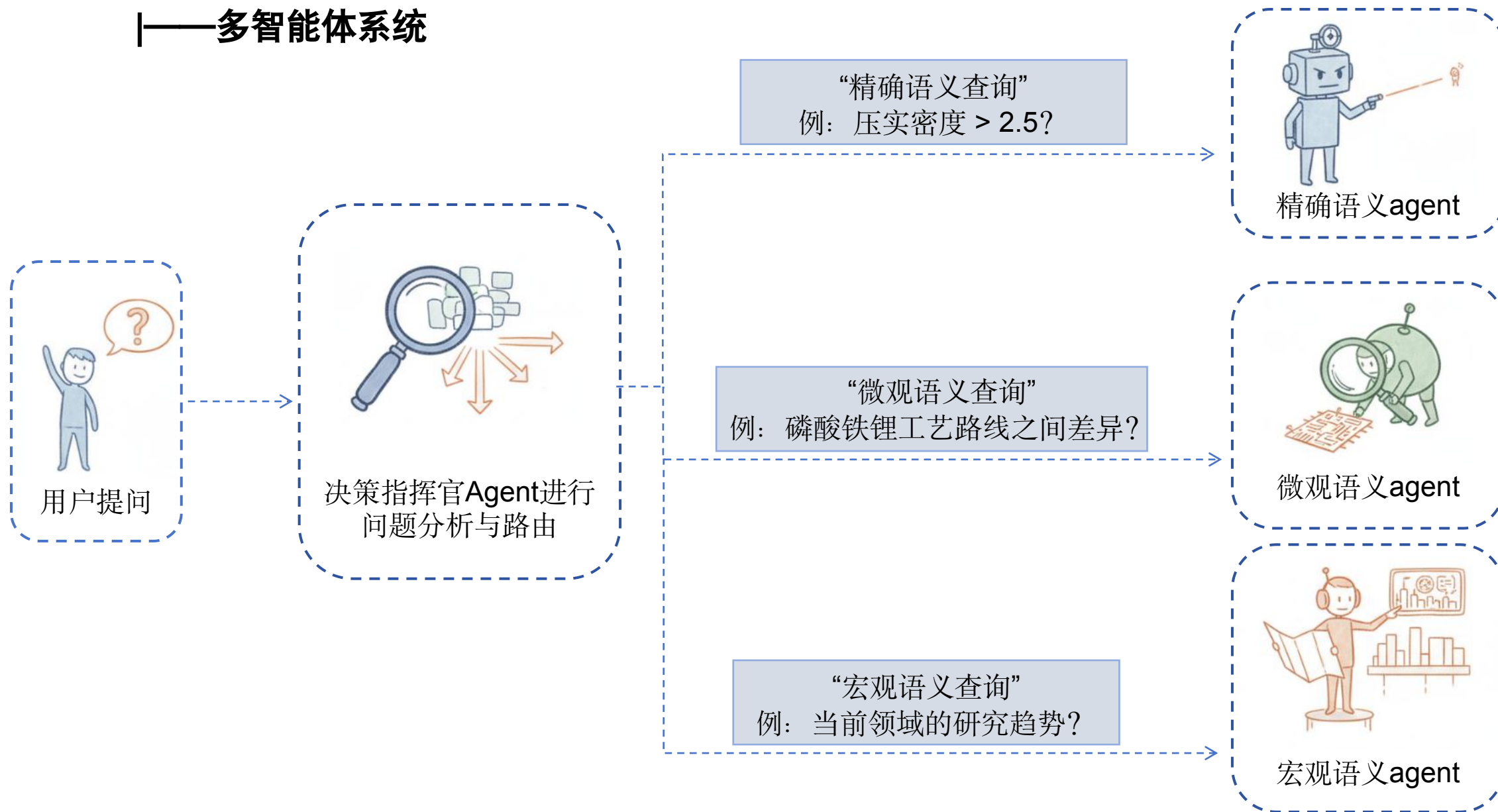
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"recipe": {
  "Fe_P_ratio": "Fe:P molar ratio = x:1 for x=0.2,0.4,0.6,0.8",
  "Li_Fe_ratio": "Li:Fe molar ratio = (1-x):x for
x=0.2,0.4,0.6,0.8",
  "other_ratios": "Precursor solutions: LiCl and FeCl3·6H2O
each 0.2 M, Na2HPO4 0.1 M",
```


03 | 多智能体问答系统的搭建



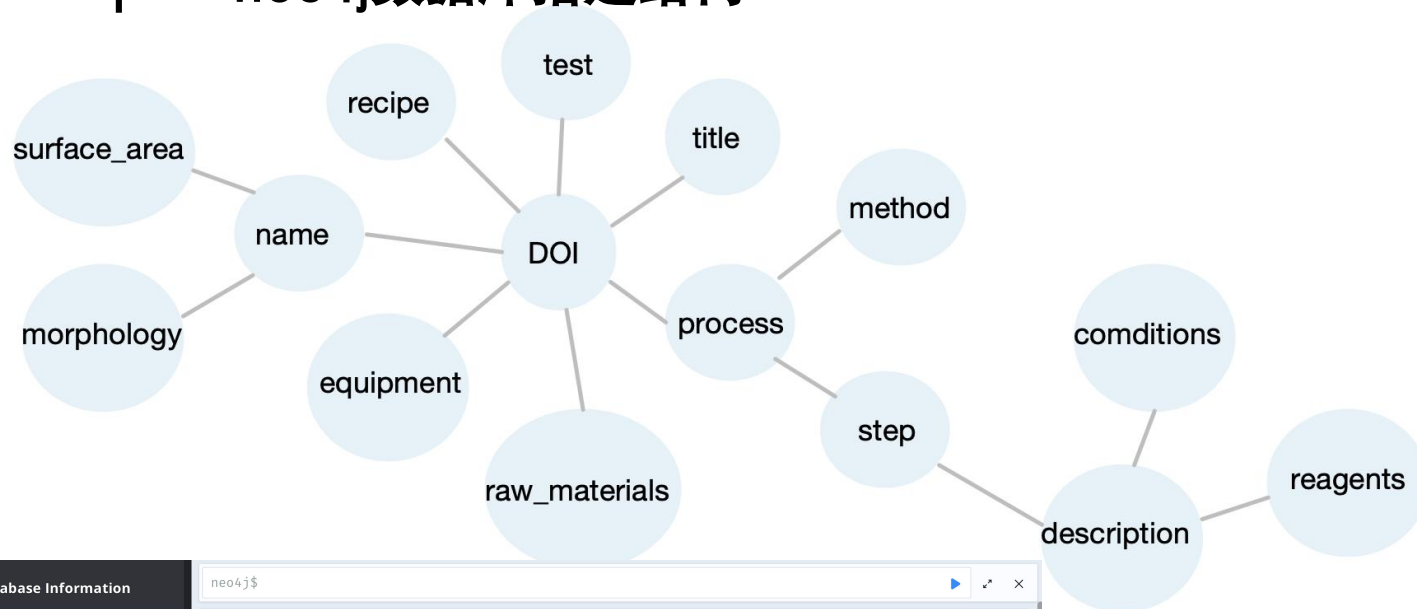
03 多智能体问答系统的搭建

——多智能体系统

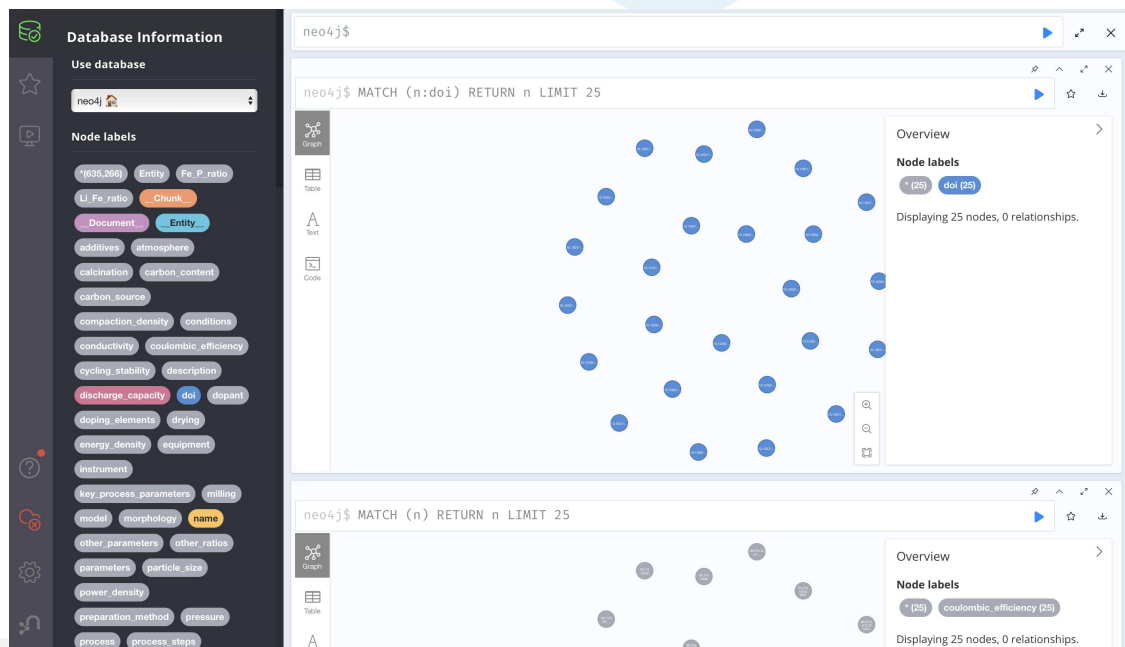


03 多智能体问答系统的搭建

——neo4j数据库搭建结构



现neo4j数据库
已经构建了
635266个节点,
855573条关系。



——向量数据库

结构化信息+文献摘要

一级	二级
原料	名称、生产厂家、纯度
配方	铁磷比、锂铁比、元素掺杂、碳源、添加剂
工艺	砂磨喷雾粉碎参数、烧结制度、工艺流程、工艺路线
性能	压实密度、形貌（圆润度）、碳含量、粉末电阻率、粒度、容量、能效
设备	名称、型号、构造、参数、产能
测试	测试方案、仪器名称

ABSTRACT

The present work focuses on developing carbon-coated lithium iron phosphate (LFP/C) cathode material from economical, high-dense Fe₃O₄ iron precursor resulting in good capacity, high atomic economy, and appreciable tap density for lithium-ion battery (LIB) fabrication. In addition, Fe₂O₃ and Fe were also used as Fe precursor for LFP/C synthesis and the results were compared with Fe₃O₄-LFP/C. Less gas evolution during heating and high-density precursors resulted in ~20–25% more atomic efficiency and ~1.5 to 1.8 times higher tap density over FeC₂O₄-LFP/C. Further, the synthesized materials have been characterized for their phase purity, morphology, and oxidation states using various characterization techniques. Electrochemical studies showed that Fe₃O₄-LFP/C delivers a high capacity of 137 mAh g⁻¹ at 1C when compared with Fe₂O₃-LFP/C and Fe-LFP/C. Interestingly, Fe₃O₄-LFP/C retained 83 % capacity after 600 cycles at 1C, illustrating the long cyclic stability. In addition, Fe₃O₄-LFP/C with the high atomic economy (73%) and tap density equivalent to the commercially available LFP/C paves the path for affordable LFP synthesis for high-energy density batteries. Hence, the LFP/C developed in this work can be used as a cathode material for high-density electrodes that are appropriate for high-energy applications.

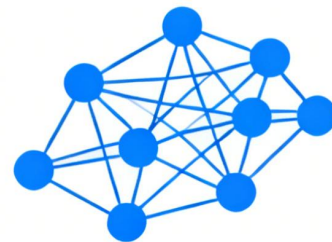
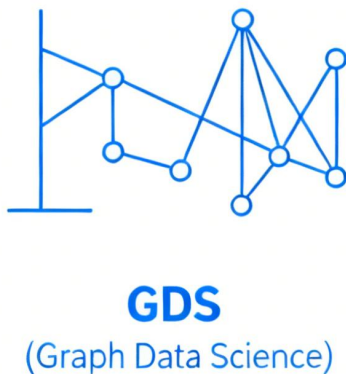
【DOI:10.1016/j.est.2023.108453】本研究基于低成本、高密度的Fe₃ O₄ 铁前驱体开发碳包覆磷酸铁锂 (LFP/C) 正极材料，旨在获得高容量、高原子经济性和优异振实密度的锂离子电池阴极材料。制备工艺采用化学计量比 Li:Fe:P = 1.05:1:1（锂过量5%），以蔗糖为碳源（10wt%），通过球磨（400 rpm，6小时）实现均匀混合，随后在80℃干燥12小时，压片（5吨压力）后于氩气氛中750℃煅烧10小时（升温速率5℃/min）。所得材料呈现均匀分布的小尺寸颗粒（平均粒径0.22 μm），具有扭曲球形形貌且无团聚现象，形成核壳结构（无定形碳层厚度10.5 nm），振实密度达1.1 g cc⁻¹，电导率提升至1.304×10⁻¹ S cm⁻¹。电化学性能显示：0.2C初始放电容量143 mAh g⁻¹，1C为137 mAh g⁻¹，2C为125 mAh g⁻¹，5C仍保持111 mAh g⁻¹；1C倍率下循环100次后容量保持130 mAh g⁻¹（容量保持率95%），600次循环后仍保持83%容量保留率，库伦效率稳定在99%。与传统方法相比，该材料通过低成本前驱体实现了高振实密度（1.1 g cc⁻¹）与优异倍率性能（5C容量达111 mAh g⁻¹）的结合，同时展现出卓越的长循环稳定性（600循环容量保持83%）。

——neo4j社区摘要数据库

实现方法：Neo4j + GDS图数据科学 + Louvain算法 + 大语言模型



图数据科学 (GDS)
能够有效地处理和分析图形数据，在进行社区检测之前，首先需要确保我们对节点与边的构成数据有充分的理解，经过 GDS 处理的图数据更加干净和结构化，这样 Louvain 算法可以更准确地输出结果。



Louvain
Community or Algorithm

Louvain 算法

初始阶段：算法将每个节点视为一个独立的社区。

合并阶段：它通过分析节点之间的连接强度，将具有高链接密度的节点合并为同一个社区。

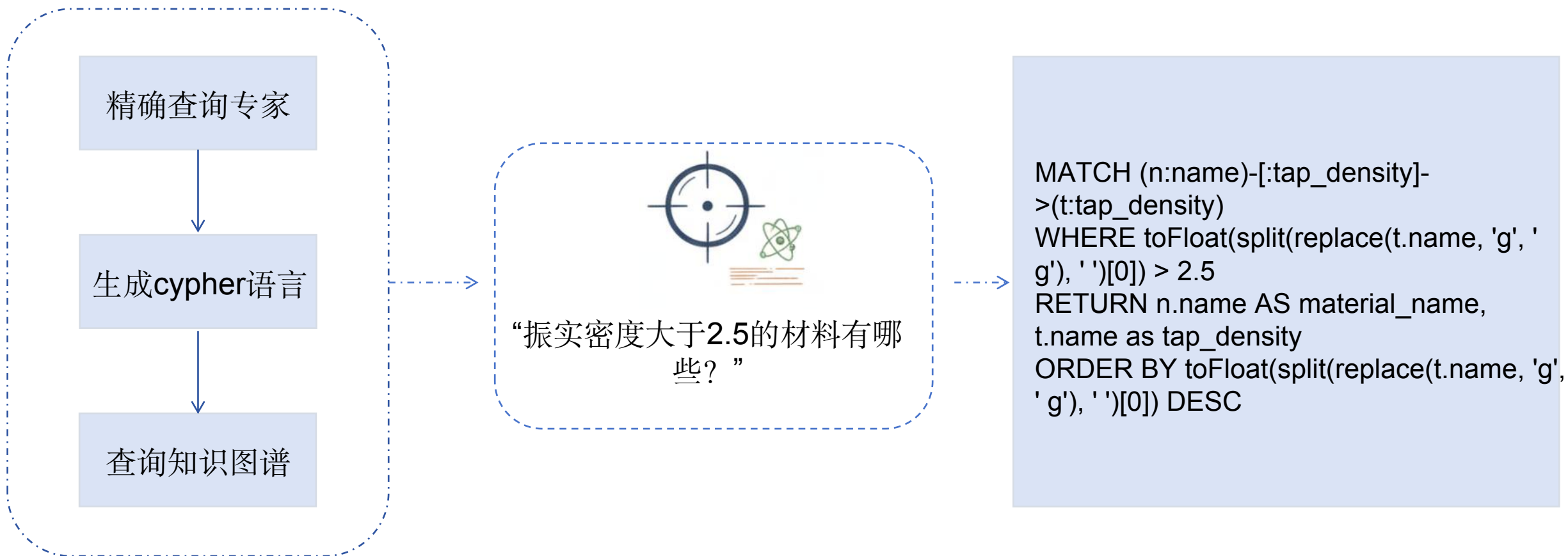
优化过程：重复合并的过程，直到无法进一步优化为止。

将处理后的社区数据提供给 LLM，并设置明确的prompt，LLM 会分析提供的信息，并生成简洁的文本摘要，描述该社区的特征、活动和主要讨论主题。



LLM
Large Language Model

——精确查询技术



——向量化匹配技术（微观与宏观语义）



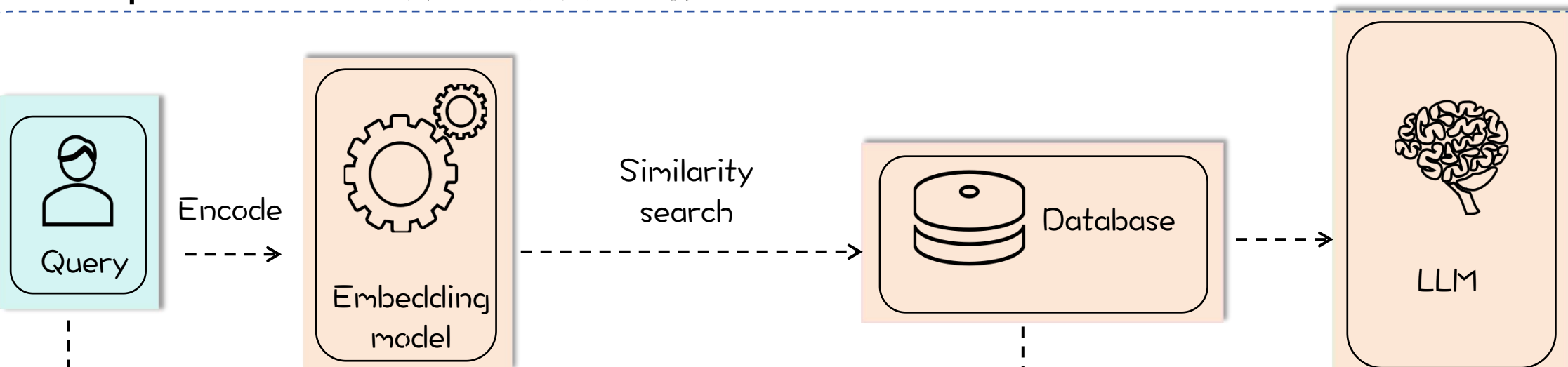
04 | 问答系统流程



04 问答系统流程

——通过DOI连接：文献原文与摘要的智能匹配

原系统简要流程



Analyse

Encode

新流程一：利用大语言模型进行用户意图分析，以实现更精准的向量匹配，从而确保获取最符合用户需求的相关数据。

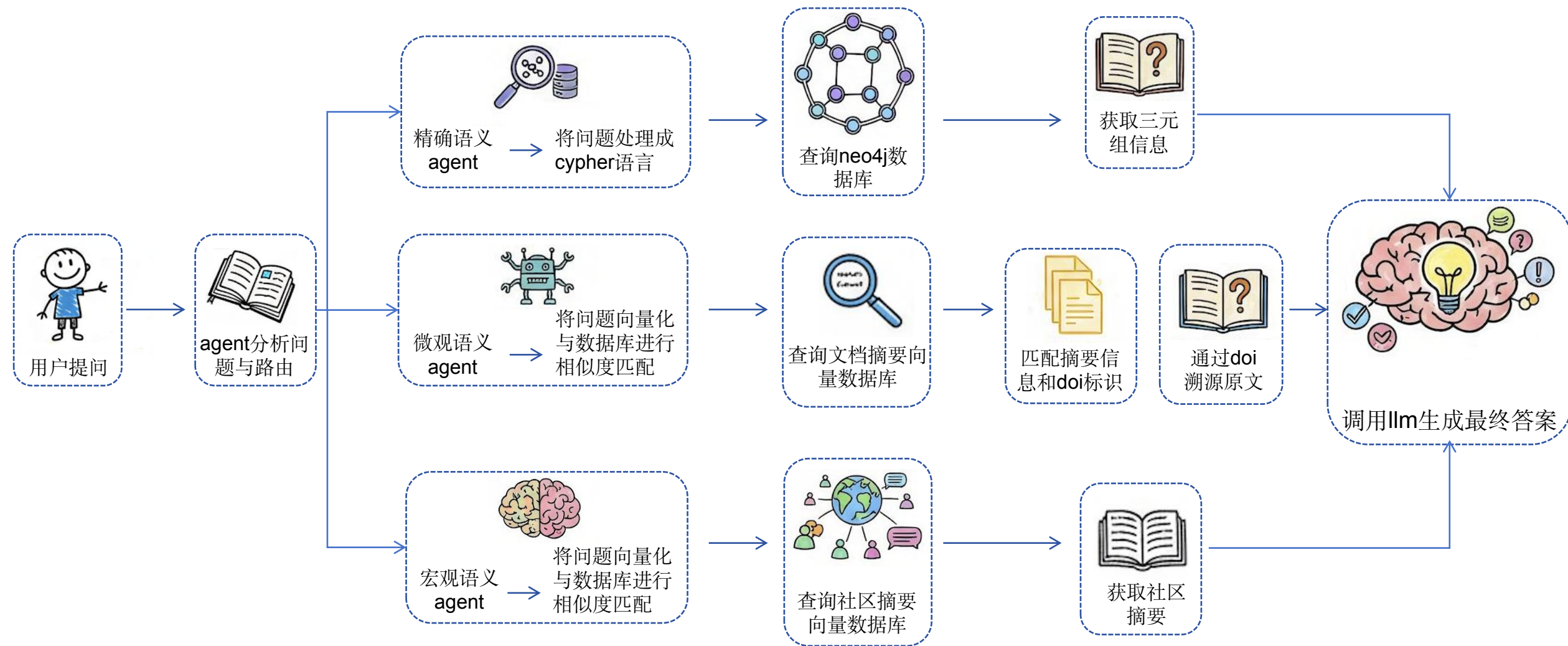
Matching

新流程二：在通过向量匹配获得摘要数据库中最相关的数据后，系统将根据每个数据的唯一DOI标识进行原文匹配，再将这些信息一并提供给大模型，以实现更加准确和专业的回答。

Documents

04 问答系统流程

——完整技术路线



|——功能升级：文献溯源· 流式呈现· 多轮对话· 历史回顾

根据后续测试反馈，已新增以下功能：

- 1.查看生成答案的原文文献
- 2.流式输出回答功能（即无需等待全部答案生成，用户可实时看到内容逐句 / 逐段呈现的过程）以提升用户体验、
- 3.具备多轮对话能力（可记忆历史沟通内容），助力更深入的沟通，
- 4.历史记录查看功能以便用户回顾之前的对话内容。
- 5.选中文献段落，精准翻译功能

04 问答系统流程

|——功能升级：翻译功能展示

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Abstract

The production and sales of lithium-ion batteries (LIB) are rapidly expanding nowadays, causing a significant impact on the consumption of critical raw materials, such as lithium. Thus, developing and improving methods for the separation and recovery of materials from LIBs is necessary to ensure the supply of critical raw materials, as well as to meet the recycling targets set by some countries. This study evaluated and compared two mechanical routes to concentrate materials of LiFePO_4 (LFP) cells. In addition, the economic, environmental, and scarcity risk potential of the products obtained through the best mechanical route were evaluated. The first route involved 6 grinding cycles in a knife mill, followed by particle size separation into 3 fractions. The second route involved a single grinding cycle (knife and hammer mill were tested), followed by particle size separation into 6 fractions. The second route showed more promise, with obtaining fractions rich in (1) iron, (2) aluminum and copper, and (3) cathode materials. Additionally, less operating time and energy consumption were necessary. The hammer mill offered a better separation for the iron and the cathodic materials (LiFePO_4), while the knife mill proved to be more effective in concentrating the aluminum and copper. The product potential evaluation of the best route revealed that the priority fractions for recycling in economic and environmental assessment in LFP2 are $2 < n < 9.5$ mm (due Cu and Al) and $n < 0.5$ mm (due Li). Considering the scarcity risk, priority should be assigned to the recycling of the fraction $n < 0.5$ due to lithium.

Keywords Spent Lithium-Ion Batteries · Mechanical Process · LFP Cell · Economic Potential · Environmental Impact · Scarcity

Introduction

Electric vehicles (EV) have many advantages compared to conventional vehicles, such as more energy efficiency, absence of greenhouse gas emissions, and noise reduction, which contribute to improving the quality of life and health of people living in urban zones. Additionally, they do not use

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fossil fuels, reducing oil dependence (Carvalho et al. 2023). The global electric vehicle market has shown exponential growth in recent years. In 2022, 14% of vehicles sold were electric, in contrast to 9% in 2021 and, 5% in 2020. China (with 60% of total sales), Europe, and the United States were the main markets (IEA 2023).

Lithium-ion batteries (LIB) are a critical component for EVs. Thus, the manufacture of this component is also expanding. In 2022, the LIB's global demand for use in EVs increased by about 65%, further increasing the consumption of critical minerals used in their manufacture. In 2022, around 60% of lithium demand, 30% cobalt, and 10% nickel were used in the production of batteries for electric vehicles (IEA 2023).

LIBs are composed of a cathode, an anode, an organic electrolyte, polymeric separators (Zhang et al. 2014), and other inactive components such as the current collector.

中文

锂离子电池 (LIB) 的生产和销售规模正在迅速扩大, 这对锂等关键原材料的消耗产生了重大影响。因此, 开发和改进从锂离子电池中分离和回收材料的方法对于确保关键原材料的供应以及实现一些国家设定的回收目标至关重要。本研究评估并比较了两种浓缩磷酸铁锂 (LFP) 电池材料的机械处理路线, 并对最佳机械路线所得产品的经济性、环境效益及稀缺风险潜力进行了评估。第一条路线采用刀式粉碎机进行6次破碎循环, 随后按粒度分离为3个组分; 第二条路线采用单次破碎循环 (测试了刀式和锤式粉碎机), 随后按粒度分离为6个组分。第二条路线展现出更好前景, 可获得富含(1)铁、(2)铝和铜、(3)正



粘贴并翻译

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翻译

05 | 后续计划



|——现已完成节点及后续工作

工作任务	任务细化	预计完成时间	当前进展
知识库构建	对15000英文文献进行信息提取	9.22-10.9	已完成
	下载中文文献	9.30-11.15	已完成
	对中文文献进行信息提取	11.16-12.2	进行中
	下载专利	11.15-12.15	进行中
	对专利进行信息提取	12.15-12.29	进行中
构建知识图谱	设计图谱结构，搭建三元组	11.13-11.26	已完成
构建问答系统	原文pdf文献翻译功能	11.19-12.10	已完成
	设计agent管理知识库	11.20-12.29	进行中
	设计召回和重排检索系统	11.20-12.29	进行中